4.1 Communicable diseases, disease prevention and the immune system

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Organisms are surrounded by pathogens and have evolved defences against them. Medical intervention can be used to support these natural defences.

The mammalian immune system is introduced.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the different types of pathogen that can cause communicable diseases in plants and animals	To include: • bacteria – tuberculosis (TB), ring rot (potatoes) • viruses – HIV/AIDS (human), influenza (animals), Tobacco Mosaic Virus (plants) • protoctista – malaria, potato/tomato late blight • fungi – black sigatoka (bananas), athlete's foot (humans). Learners are **not** required to know the binomial name of the pathogens that cause the diseases listed above.

(b)	the means of transmission of animal and plant communicable pathogens	To include an understanding of the different methods of transmission with reference to vectors, spores and living conditions – e.g. climate, social factors (no detail of the symptoms of specific diseases is required). Learners are **not** required to categorise pathogens between direct and indirect methods of transmission. *M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.5, M1.7, M3.1, M3.2* HSW1, HSW2, HSW3, HSW5, HSW6, HSW7, HSW8, HSW11, HSW12
(c)	plant defences against pathogens	To include production of chemicals **AND** plant responses that limit the spread of the pathogen (e.g. callose deposition).
(d)	the primary non-specific defences against pathogens in animals	Non-specific defences to include skin, blood clotting (limited to platelets releasing substances that, via a cascade of events, result in the formation of fibrin which itself forms a network, trapping platelets and forming a clot), wound repair, inflammation, expulsive reflexes and mucous membranes. Learners are **not** required to recall names of clotting factors or all steps of the clotting cascade. Learners are **not** required to recall details of skin structure. HSW2, HSW8
(e)	**(i)** the structure and mode of action of phagocytes **(ii)** examination and drawing of cells observed in blood smears	To include neutrophils and antigen-presenting cells **AND** the roles of cytokines, opsonins, phagosomes and lysosomes. **PAG1** HSW4, HSW8
(f)	the structure, different roles and modes of action of B and T lymphocytes in the specific immune response	To include the significance of cell signalling (reference to interleukins), clonal selection and clonal expansion, plasma cells, T helper cells and T killer cells. HSW8
(g)	the primary and secondary immune responses	To include T memory cells and B memory cells. *M1.3* HSW2

(h)	the structure and general functions of antibodies	To include the general protein structure of an antibody molecule.
(i)	an outline of the action of opsonins, agglutinins and anti-toxins	
(j)	the differences between active and passive immunity, and between natural and artificial immunity	To include examples of each type of immunity.
(k)	autoimmune diseases	To include an appreciation of the term *autoimmune disease* and arthritis as a named example.
(l)	the principles of vaccination and the role of vaccination programmes in the prevention of epidemics	To include routine vaccinations **AND** reasons for changes to vaccines and vaccination programmes (including global issues). *M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.5, M1.7, M3.1, M3.2* HSW1, HSW2, HSW3, HSW5, HSW6, HSW7, HSW8, HSW9, HSW11, HSW12
(m)	possible sources of medicines	To include examples of microorganisms and plants (and so the need to maintain biodiversity) **AND** the potential for personalised medicines. HSW7, HSW9, HSW11, HSW12
(n)	the benefits and risks of using antibiotics to manage bacterial infection.	To include the wide use of antibiotics following the discovery of penicillin in the mid-20th century **AND** the increase in bacterial resistance to antibiotics and its implications. HSW2, HSW5, HSW9, HSW12